

packages/graph-view/src/hooks/useEmbedAutoExpand.ts

Kind: added · Baseline: a9c34bfb03fe · Target: NovaDiff · Generated: 2026-05-20T05:32:59.117Z

Overview

A new hook, `useEmbedAutoExpand`, is added to `packages/graph-view/src/hooks/useEmbedAutoExpand.ts` (lines R42-129). It expands folder containers and fits the viewport when a NovaDiff embed is active.

Key changes

- **Imports** (R1-R4) ``ts import { useEffect, useRef, useState } from "react"; import { useReactFlow } from "@xyflow/react"; import { useDashboardStore } from "../store"; import { useNovaDiffEmbed } from "../contexts/NovaDiffEmbedContext";

```
`EXPAND_POLL_MS = 100` ms, `EXPAND_TIMEOUT_MS = 120 000` ms.
- Interface (R9-R15)
  `EmbedAutoExpandInput` defines navigation level, active layer, layout flag, container IDs, and
  graph fingerprint.
- Helper functions
  - `containersFullyLaidOut` (R17-R21) checks `expandedContainers` and `containerLayoutCache`.
  - `waitForContainerLayouts` (R24-R35) polls every `EXPAND_POLL_MS` until all target containers
  are laid out or the timeout expires.
- Hook logic (R42-R129)

* Activates only when `embedMode` is true and the navigation level matches (`overview` or `layer-
detail`).

* Expands missing containers via `expandManyContainers`.

* Waits for layout with `waitForContainerLayouts`, then calls `fitView` (duration 550 ms,
padding 0.28, maxZoom 1, minZoom 0.2).

* Uses `runKeyRef` to avoid duplicate work and `cancelledRef` to abort pending operations.

* Returns `isExpanding` to indicate ongoing expansion.

### Impact
```

- **Correctness** – ensures containers are expanded before `fitView` is invoked.
- **Maintainability** – centralizes embed-specific logic; helper functions are pure and testable.
- **Performance** – polls every 100 ms up to a 120 s timeout; bounded by the defined constants.

- **Compatibility** – relies on existing `useDashboardStore` and `useReactFlow`; no public API changes.

Risks & follow-ups

- **Race conditions** – `waitForContainerLayouts` may not handle rapid layout changes; test with concurrent expansions.
- **Timeout edge cases** – 120 s may be excessive for very large graphs; consider exposing the timeout.
- **Embed mode detection** – depends on `useNovaDiffEmbed`; verify it reports `embedMode` correctly.
- **Viewport fitting** – validate that the `fitView` parameters produce the intended zoom/center across screen sizes; adjust padding if needed.